

Age Estimation from Speech

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Abstract—This report focuses on developing a regression model to estimate individuals' ages using a comprehensive set of features, including audio characteristics and demographic features. A comparison of various regression models identified Random Forest as the most effective for capturing complex relationships. Despite challenges such as dataset imbalance, the model achieved competitive results, highlighting the importance of robust preprocessing and feature engineering.

I. PROBLEM OVERVIEW

This analysis addresses the problem of *age estimation* from speech data. The dataset contains 3,624 samples, divided into:

- Development set: 2,933 samples
- Evaluation set: 691 samples

Each sample represents a spoken sentence recorded under controlled conditions, with the speaker's age as the target label, which is represented as a continuous numerical value, making this suitable for regression analysis. The dataset provides a range of features, including:

- Audio characteristics: sampling rate, pitch measurements (mean, max, min), jitter, shimmer, energy, harmonic-to-noise ratio (HNR), zero-crossing rate (ZCR), spectral centroid;
- Linguistic features: number of words, characters, pauses, and silence duration;
- Demographic informations: gender, ethnicity.

Additionally, there are the raw audio recordings, of both development and evaluation set, allowing the extraction of additional acoustic characteristics.

We can draw some conclusions from the development set. First, the audio signals are consistently sampled at a rate of 22050 Hz. However, the durations of the recordings vary significantly, spanning from a maximum of 106.54 seconds to a minimum of 0.67 seconds, with an average length of 16.62 seconds. Most of the samples have durations of less than 5 seconds or fall between 20 and 30 seconds.

The dataset is imbalanced, as the majority of samples fall within the *age* range of 20 to 30 years, leaving only a small fraction of samples that are either younger or older, as evidenced by the asymmetric distribution of the variable. The target variable *age*, in fact, exhibits a right-skewed distribution with a skewness equal to 1.79 and a kurtosis of 3.16, indicating a slightly asymmetric and sharp distribution compared to a normal one. From the graph in *Figure 1* we can see that most of the observations are concentrated in younger age groups,

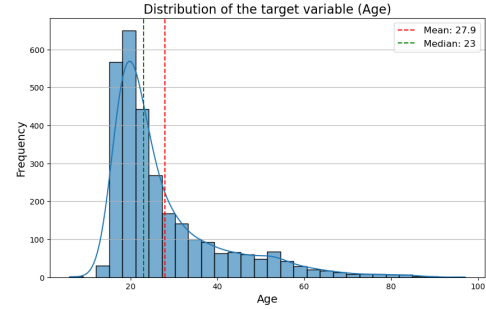


Fig. 1: Age distribution

with a mean of 27.9 years and a median of 23 years. Due to this distinction we expect to have a model more suitable for younger individuals.

To gain a better understanding of the audio characteristics within our dataset, we conducted a visual analysis of the audio signals.

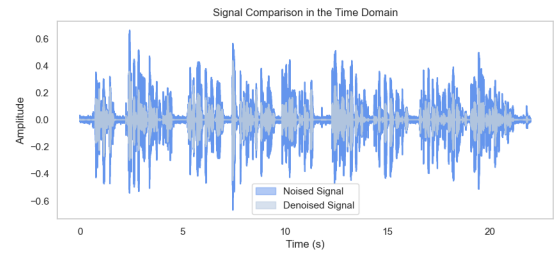


Fig. 2: Time domain representation of the audio signal n_{audio}

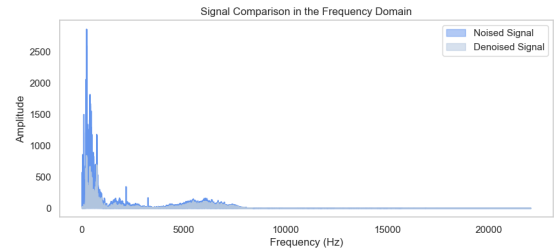


Fig. 3: Frequency domain representation (linear scale) of the audio signal n_{audio}

The Time domain Representation illustrates the amplitude of the signal as it evolves over time, offering valuable insights into the temporal characteristics of speech. In Figure 2, the

From the analysis of the audio, it has emerged that there are no pauses at the beginning and at the end of the recordings; however, the audio is not defined very well. Therefore, a noise reduction process was applied to the recordings using the *noisereduce* package [1], which is also evident from the visual analysis. In the Frequency domain (Figure 3), it can be observed a reduction in the low frequencies typically associated with noise, while in the Time domain (Figure 2), the silent parts (where the wave is close to zero) are more evident, without unnecessary disturbances or fluctuations.

During the model training process, feature engineering was applied multiple times, creating new variables to enhance model performance. The first variable, *pitch_range*, measures the squared range of pitch, quantifying variability between maximum and minimum values. The second variable, *voice_stability*, combines *jitter* and *shimmer* values for both genders to assess average voice stability, providing insight into overall consistency and quality. From the analysis of the major errors made by the model, it emerged that the model had difficulty determining individuals over 40 years old, due to class imbalance. The study indicated that the *hnr* (Harmonic-to-Noise Ratio) for these individuals was below average. To give greater importance to these individuals, we created a new

variable that emphasizes the difference between their hnr value and the average of the feature.

From the analysis of several documents, it emerged that both the first 12th Mel-frequency cepstral coefficients (MFCC) and the primary formants, are important for the age determination from speech signals [2], [3]. Therefore, we decided to utilize audio data to extract these additional variables.

MFCCs provide a compact and effective representation of signal's acoustic features. They are widely used in speech recognition systems and are based on the human auditory system, which perceives frequency content non-linearly [4]. The calculation of MFCCs involves:

- 1) A filter is applied to emphasize higher frequencies and improve the signal-to-noise ratio.
- 2) The continuous signal is divided into overlapping frames to analyze local time-varying features.
- 3) A discrete Fourier transform is performed on each frame to obtain the power spectrum.
- 4) The human perception of frequency is simulated by mapping linear frequencies to the Mel scale.

$$\text{Mel}(f) = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right)$$

A filter bank with triangular bandpass filters is applied, with spacing and bandwidth determined by the Mel scale.

- 5) The logarithm of the energy for each Mel filter is computed to compress the dynamic range and reduce the effect of variations.
- 6) The log Mel spectrum is transformed back to the time domain using the discrete cosine transform to obtain the cepstral coefficients.

$$C_n = \sum_{k=1}^K \log(S_k) \cdot \cos\left[n \cdot \frac{\pi}{K} \cdot (k - 0.5)\right]$$

where S_k is the Mel spectrum energy, n is the cepstral coefficient index, and K is the number of Mel filters.

The MFCCs related to the audio signal n_{audio} are plotted in Figure 6. In this plot, for each of the 12 coefficients, how energy varies over time is shown, represented by different colors and intensities.

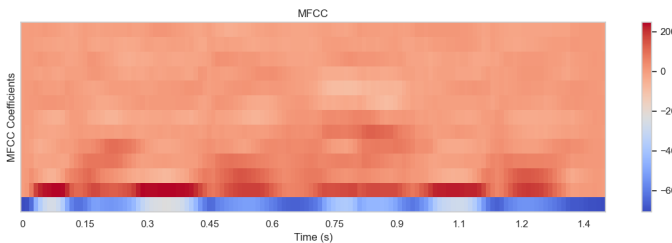


Fig. 6: MFCC of the audio signal n_{audio}

Formant frequencies are the amplitude peaks within the spectral structure of a sound, defining its characteristic qualities. Formants are particularly useful for capturing the physiological and acoustic properties of speech, making them

valuable for tasks such as age estimation [5]. Various methodologies exist for estimating these features. In this study, we compute the averaged spectrum derived from the Short-Time Fourier Transform (STFT) and subsequently identify the peaks within this spectrum, which correspond to the formant frequencies. After evaluation, we decided to retain only the first formant (F1) for further analysis, as including higher-order formants introduced redundancy without significantly enhancing model performance.

B. Model selection

To choose the best model to estimate the age we started comparing three regression models: Linear Regression, Random Forest, and Support Vector Regression. We used Grid Search to find the best parameters configuration for each model and compared them using R^2 and RMSE (Table I).

Model	Best Parameters	R^2	RMSE
Linear Regression	{}	0.2990	9.1789
Random Forest	'max_depth': 20, 'min_samples_split': 5, 'n_estimators': 100	0.2876	9.2535
Support Vector Regression	'C': 10, 'gamma': 'scale', 'kernel': 'linear'	0.2100	9.7446

TABLE I: Comparison of Regression Models

The results indicated that Linear Regression achieved the highest R^2 and the lowest RMSE, closely followed by Random Forest, while Support Vector Regression resulted to be the worst.

However, the feature importance analysis for Linear Regression highlighted that the model relies heavily on variables such as:

jitter_male: 204.8593, *continent_South_America*: 18.3114,
jitter_female: 135.7703, *continent_North_America*: 10.6465,
shimmer_female: 19.525067

These variables are intrinsically linked to gender and ethnicity and alone are not appropriate predictors of age because they will not reflect causal relationships and will introduce bias.

While both models achieve comparable performance, Random Forest focuses on an enriched and broader set of features, making it a more balanced and robust choice for the task. Further optimization of the Random Forest model was conducted using Grid Search to fine-tune parameters such as *max_depth*, *min_samples_leaf*, and *n_estimators*. To enhance predictions, additional features, including *pitch_range* and *voice_stability*, were introduced. Moreover, a new feature, *hnr_diff*, was added to better capture voice quality variations related to aging.

We conducted a comparison of the correlations obtained from Pearson and Spearman analyses, highlighting the presence of nonlinear relationships with age. We considered increasing the degree of the polynomial features to better capture the complexity of the interactions between the variables, improving

performance in cases where the relationship between variables is not strictly linear.

C. Hyperparameters tuning

Following the evaluations in the previous section, a pipeline was created using polynomial features and a Random Forest Regressor. The hyperparameter tuning aimed to identify the combination that maximized model performance while preventing overfitting. We utilized Grid Search to systematically optimize the hyperparameters, which are detailed in Table II.

Model	Parameter
Preprocessing	Polynomial Features: degree {1, 2}
Random Forest	$n_{\text{estimators}}$: {100, 200, 300} max depth: {4, 6, 8} min samples split: {2, 5, 10} min samples leaf: {1, 2, 5} max features: {None, sqrt, log2} bootstrap: {True, False}

TABLE II: Model Parameter Values

We allocated 85% of the development set for training and 15% for testing, as this approach led us to better results. This change allows the model to train on a larger portion of data, having more examples from underrepresented age groups, which enhanced its ability to learn and perform better for those specific demographics.

III. RESULTS

The best model resulted to be the Random Forest with an RMSE of 6.10 on the training set, 8.11 on the test set and 9.770 on the evaluation set and a R^2 equal to 0.326. The best configuration of hyperparameters, obtained through Grid Search, is represented in Table III.

Parameter	Value
degree	2
bootstrap	False
max depth	8
max features	sqrt
min samples leaf	5
min samples split	10
n estimators	200

TABLE III: Hyperparameters Configuration

To evaluate if the model performs better than a random guess, we compared its performance to that of a random prediction model. For this comparison, we assumed that the ages in the dataset followed a normal distribution, using the mean and standard deviation of the actual age values of the training set. We obtained a R^2 equal to -0.9438 and a RMSE of 13.9068.

To compare the results to a naïve solution we computed R^2 and RMSE of a model that predict the mean age for all samples, obtaining respectively -0.0149 and 10.0486. The

Random Forest model outperforms both the random guess and the naive models having an higher R^2 and a lower RMSE.

The solution secured a respectable position (103 out of 180) on the public leaderboard with a score of 9.770. The better score among participants was 8.961, while the worst was 10.687. Considering that traditional approaches were used without implementing neural networks or machine learning, this score can be regarded as acceptable.

IV. DISCUSSION

The results indicate that hyperparameter tuning and the inclusion of polynomial features improved the model ability to capture non-linear relationships. However, the higher RMSE on test data compared to training data suggests the model may still face challenges with generalization, particularly for underrepresented age groups.

When comparing the true values with the predicted values on the test set, it is evident that the model performs better at predicting ages under 40, but struggles with estimating ages above 40. As can be seen in the graph in Figure 7, for true values above 40-60, the model appears to underestimate the predictions, creating a "curve" in the graph.

This discrepancy is likely due to the under-representation of older age groups in the training set, which the same graph highlights, emphasizing the imbalanced nature of the problem. To address this, we tested different balancing strategies such as oversampling, adjusting weights assigned to the different age groups and bin partitioning. However, neither of these approaches lead to significant improvement, highlighting the need of a more balanced dataset. Future efforts should focus on collecting more data for these underrepresented age groups or identifying alternative features that better capture subtle vocal changes associated with aging. These adjustments could help ensure a fairer representation enhancing the overall predictive performance.

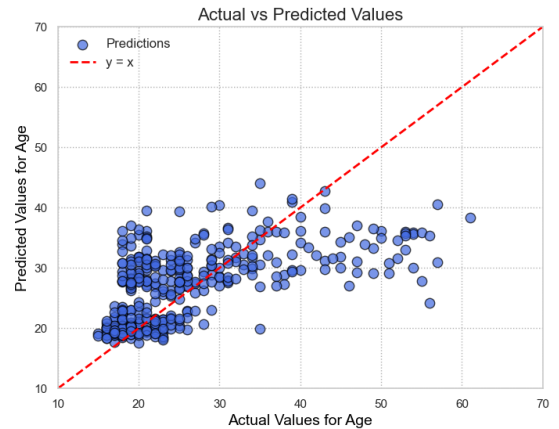


Fig. 7: Random Forest Regression: True vs Predicted Values

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